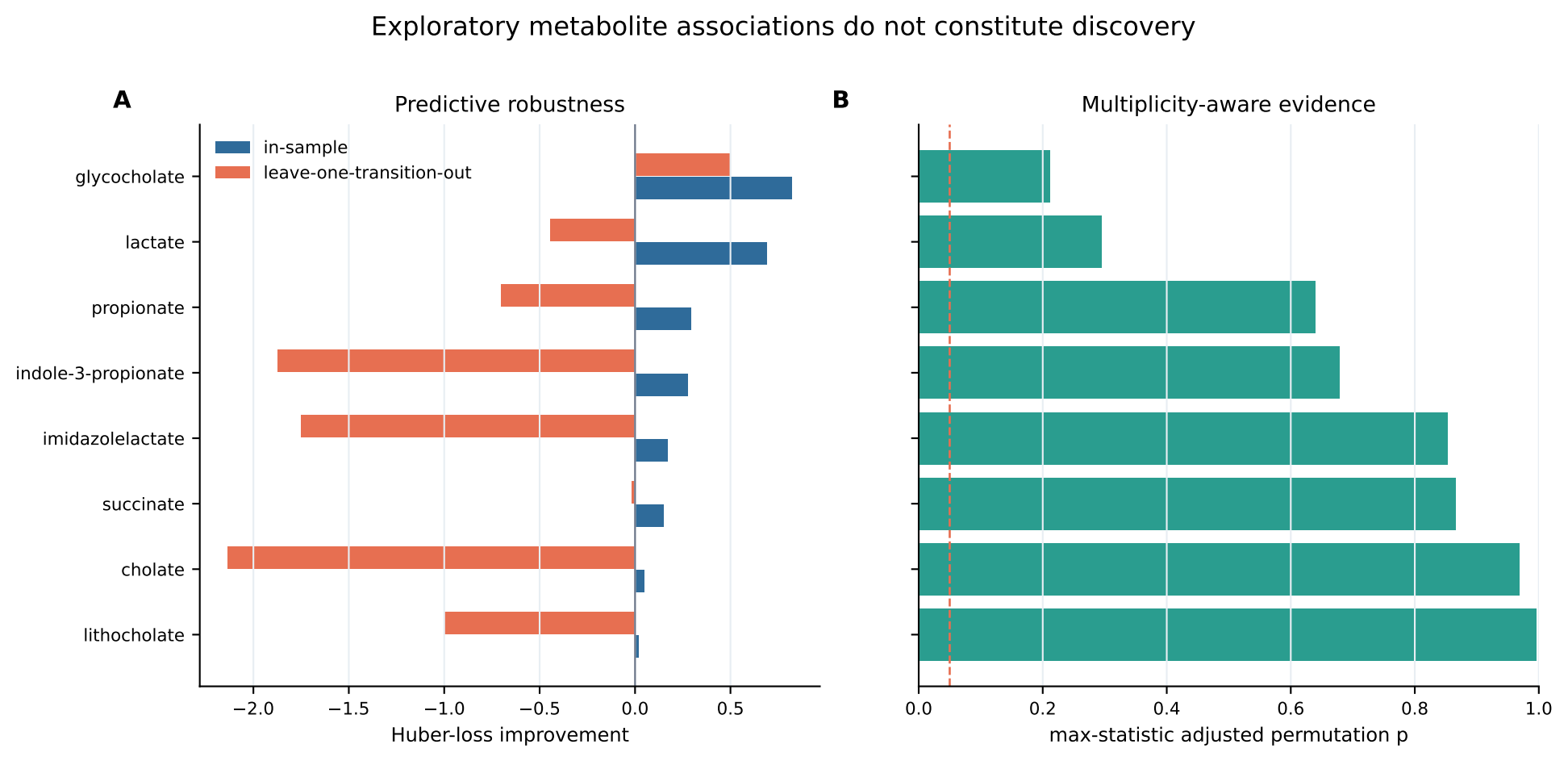
Exploratory metabolite associations do not constitute discovery
A
Predictive robustness
in-sample
leave-one-transition-out
glycocholate
lactate
propionate
indole-3-propionate
imidazolelactate
succinate
cholate
lithocholate
−2.0
−1.5
−1.0
−0.5
0.0
0.5
Huber-loss improvement
B
Multiplicity-aware evidence
0.0
0.2
0.4
0.6
0.8
1.0
max-statistic adjusted permutation p